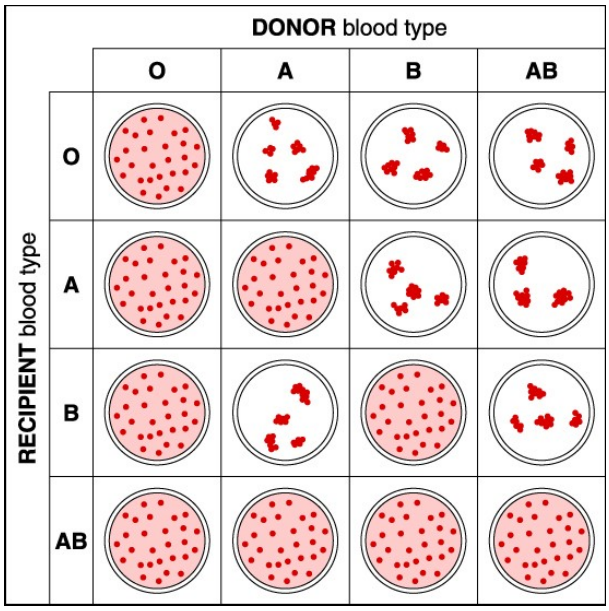
Blood transfusions between one person and another can result in hemolysis and agglutination of the red blood cells due to antigenic reactions. Although there are at least 30 different commonly occurring antigens on blood cells, the two types most likely to produce a transfusion reaction are the ABO antigens and the Rh system.

Two antigens, known as A and B can be found on the surface of the red blood cells. Because of the way in which they are genetically inherited, persons can have either A, B, both A and B, or neither of the antigens.





A patient with Type A blood have anti-B antibodies. Patients with Type B blood have anti-A antibodies. Patients with Type O blood have both anti-A antibodies and anti-B antibodies and patients with Type AB have neither.

The most common reaction from the mixing of type A blood with type B blood is the agglutination (clumping together) of the cells. Because the cells of type O blood don't contain the A or B surface antigens, it can be administered universally. Patients with type AB blood can receive any type of blood.

If the patient has a particularly high titer of antibodies to the blood they are receiving, the complement system can be activated, which results in the release of proteolytic enzymes that can produce massive hemolysis.

Another system is the Rh class of antigens, which are known as the C,c,D,d,E, and e antigens. The D antigen is the most common and capable of producing the strongest reactions. A patient who has the D antigen is referred to as Rh positive. If the D antigen is absent, they are called Rh-negative. Approximately 90% of Americans are Rh positive.

The first exposure of an Rh negative patient usually produces no reaction. Over the next 2-4 weeks, however, the patient develops enough antibodies to produce agglutination of the previously transfused cells still circulating. This is referred to as a delayed transfusion reaction. If the patient receives another transfusion of Rh positive blood, the reaction is as quick and severe as a transfusion of mismatched type A or type B blood.

Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 445-447.

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